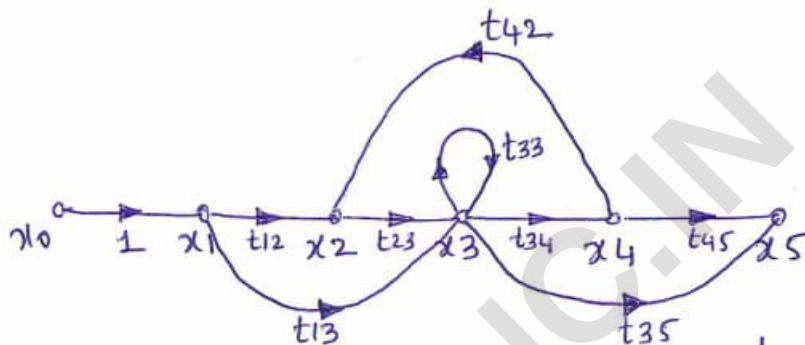


Signal Flow Graphs :

The graphical representation of the variables of a set of linear algebraic equations representing the system is called signal flow graph representation.

Terminology used in Signal Flow Graph:

Consider a signal flow graph shown below.



- (i) Source Node: The node having only outgoing branches is known as source or input node.
 Eg: x_0 is the source node.
- (ii) Sink node: The node having only incoming branches is known as sink or output node.
 Eg: x_5 is the sink node.
- (iii) Chain node: A node having incoming and outgoing branches is known as chain node.
 eg: x_1, x_2, x_3 and x_4 .
- (iv) Forward path: A path from the input to output node is defined as forward path.
 eg: $x_0 - x_1 - x_2 - x_3 - x_4 - x_5 \rightarrow$ First forward path.
 $x_0 - x_1 - x_3 - x_4 - x_5 \rightarrow$ Second forward path.
 $x_0 - x_1 - x_3 - x_5 \rightarrow$ Third forward path.
 $x_0 - x_1 - x_2 - x_3 - x_5 \rightarrow$ Fourth forward path.

(v) Feedback loop: A path which originates from a particular node and terminating at the same node, travelling through at least one other node, without leaving any node twice is called feedback loop.

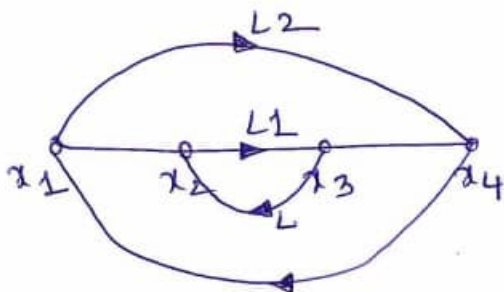
eg: $x_2 - x_3 - x_4 - x_2$

(vi) Self loop: A feedback loop consisting of only one node is called self loop, i.e. t_{33} at x_3 is a self loop.

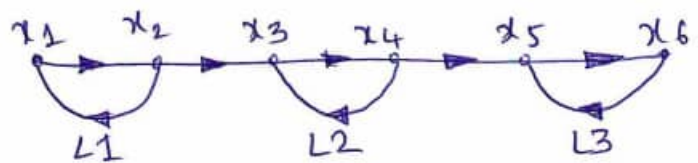
(vii) Path gain: The product of branch gains while going through a forward path is known as path gain.
eg: Path gain for the path $x_0 - x_1 - x_2 - x_3 - x_4 - x_5$ is,
 $1 \times t_{12} \times t_{23} \times t_{34} \times t_{45}$.

(viii) Non-touching loops: If there is no node common in between the two or more loops, such loops are said to be non-touching loops.

The figures show a combination of non-touching loops of two and three loops.



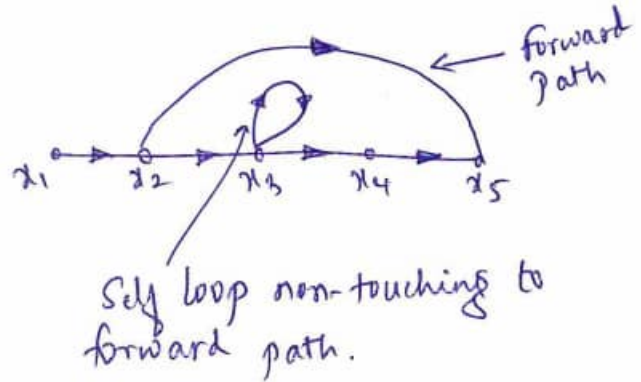
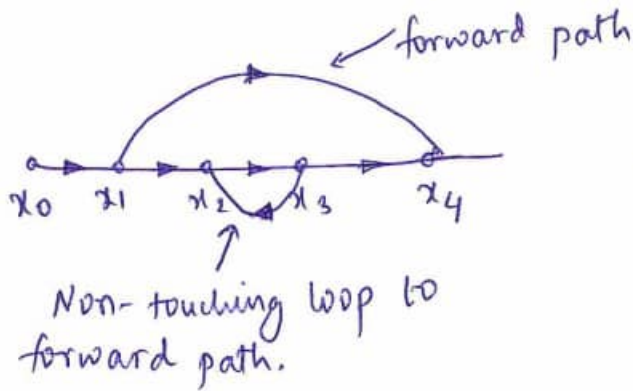
Two non-touching loops



Three non-touching loops.

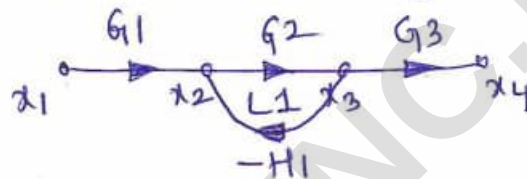
Similarly if there is no node common in between a forward path and a feedback loop, a loop is said to be non-touching to that forward path.

The following figures explain the same.

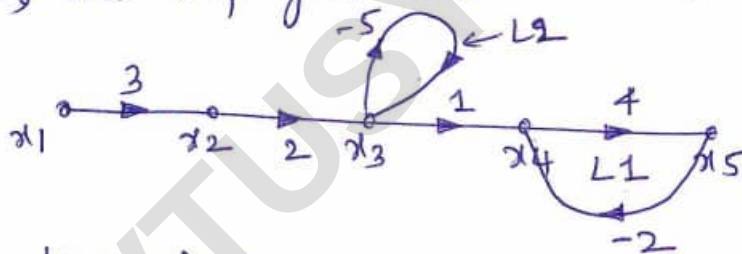


(ix) Loop Gain: The product of all the gains of the branches forming a loop is called loop gain. Generally such loop gains are denoted by 'L' i.e. L_1, L_2, L_3 etc.

Examples:



Here, the loop gain is, $L_1 = G_2 \times -H_1 = -G_2 H_1$



The loop gain $L_1 = 4 \times -2 = -8$

The loop gain of the self-loop, $L_2 = -5$

Mason's Gain Formula:

In signal flow graph approach, direct use of the formula leads to the overall system transfer function $C(s)/R(s)$. This formula is stated by Mason and hence referred to as Mason's gain formula and can be stated as,

$$\text{Overall Transfer Function} = \frac{\sum T_k \cdot \Delta_k}{\Delta}$$

where, K = Number of forward paths.

T_K = Gain of K^{th} forward path.

Δ = System determinant can be calculated as:

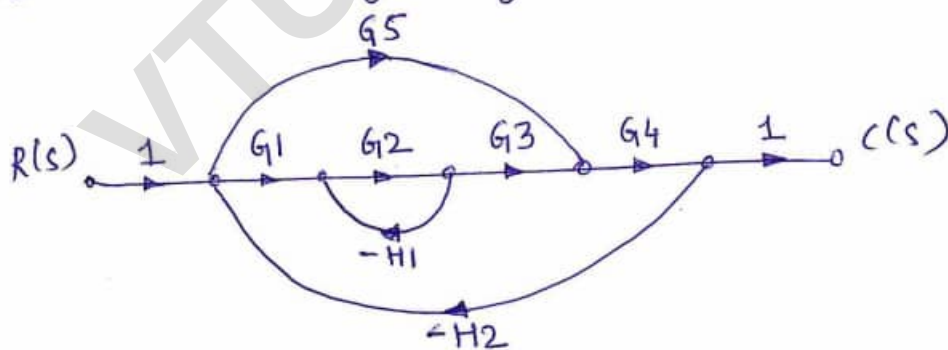
$$\Delta = 1 - [\sum \text{all individual feedback loop gains}] \\ + [\sum \text{Gain} \times \text{Gain product of all possible combinations of two non-touching loops}]$$

$$- [\sum \text{Gain} \times \text{Gain} \times \text{Gain product of combinations of three non-touching loops}] \\ + \dots$$

Δ_K = Value of above Δ by eliminating all loop gains and associated products which are touching to the K^{th} forward path.

Numericals:

① Find $C(s)/R(s)$ by using Mason's gain formula.



Solⁿ: Number of forward paths $K = 2$

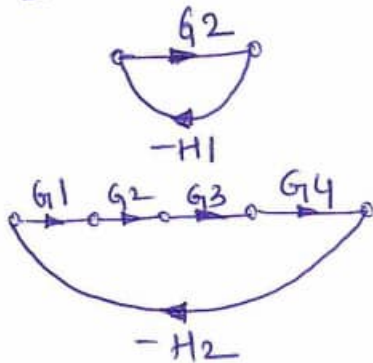
Mason's gain formula,

$$T.F = \frac{\sum_{K=1}^2 T_K \Delta_K}{\Delta}$$

$$= \frac{T_1 \Delta_1 + T_2 \Delta_2}{\Delta}$$

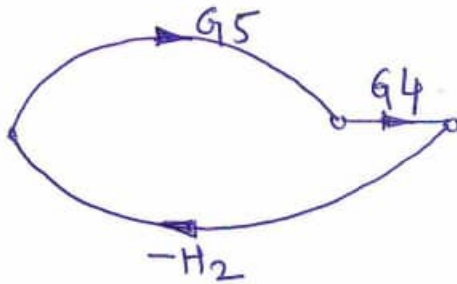
$$T_1 = G_1 G_2 G_3 G_4, T_2 = G_5 G_4$$

Individual feedback loops ~~are~~ and their gains are,



$$L_1 = -G_2 H_1$$

$$L_2 = -G_1 G_2 G_3 G_4 H_2$$



$$L_3 = -G_5 G_4 H_2$$

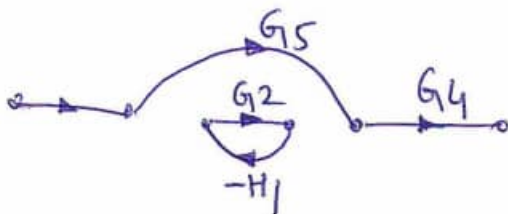
L_1 and L_3 are two non-touching loops.

$$\begin{aligned} \therefore \Delta &= 1 - [L_1 + L_2 + L_3] + [L_1 L_3] \\ &= 1 - [-G_2 H_1 - G_1 G_2 G_3 G_4 H_2 - G_5 G_4 H_2] + \\ &\quad [G_2 H_1 \cdot G_5 G_4 H_2] \\ &= 1 + G_2 H_1 + G_1 G_2 G_3 G_4 H_2 + G_5 G_4 H_2 + G_2 H_1 G_5 G_4 H_2 \end{aligned}$$

For T_1 all loops are touching.

$\therefore \Delta_1 = 1$ eliminating all loop gains and products from Δ .

For T_2 , L_1 is non-touching.



$$\Delta_2 = 1 - [L_1]$$

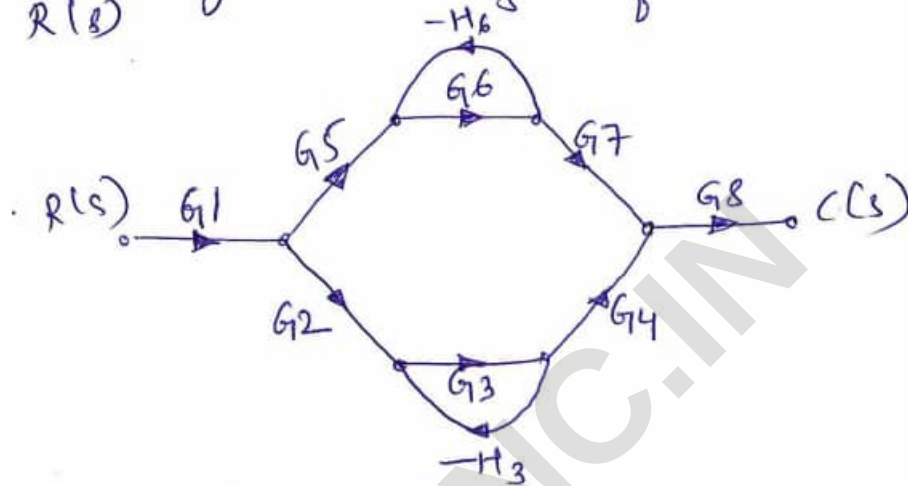
$$= 1 - [-G_2 H_1]$$

$$\Delta_2 = 1 + G_2 H_1$$

$$\therefore \frac{C(s)}{R(s)} = \frac{T_1 \Delta_1 + T_2 \Delta_2}{\Delta}$$

$$\therefore \frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4 + G_5 G_4 (1 + G_2 H_1)}{1 + G_2 H_1 + G_1 G_2 G_3 G_4 H_2 + G_5 G_4 H_2 + G_1 G_5 G_4 H_1 H_2}$$

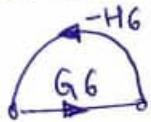
② find $\frac{C(s)}{R(s)}$ by Mason's gain formula.



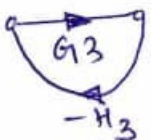
Solⁿ: Number of forward paths $\Rightarrow K = 2$

Mason's gain formula, T.F =
$$\frac{\sum_{k=1}^2 T_k \Delta_k}{\Delta} = \frac{T_1 \Delta_1 + T_2 \Delta_2}{\Delta}$$

$\therefore T_1 = G_1 G_5 G_6 G_7 G_8$ and $T_2 = G_1 G_2 G_3 G_4 G_8$
Individual feedback loops and their gains are,



$$L_1 = -G_6 H_6$$



$$L_2 = -G_3 H_3$$

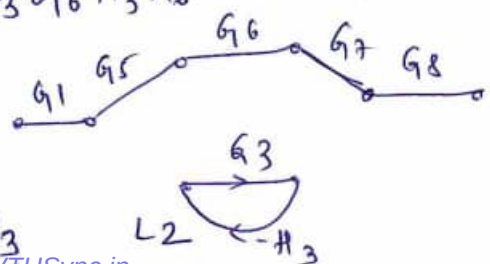
L_1 and L_2 are the two non-touching loops.

$$\therefore \Delta = 1 - [L_1 + L_2] + [L_1 L_2]$$

$$= 1 + G_6 H_6 + G_3 H_3 + G_3 G_6 H_3 H_6$$

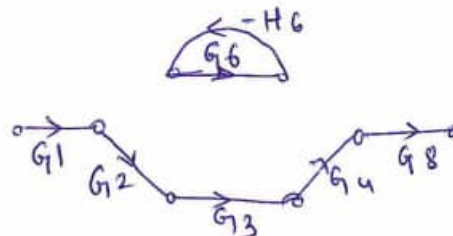
For T_1 , L_2 is non-touching

$$\therefore \Delta_1 = 1 - L_2 = 1 + G_3 H_3$$



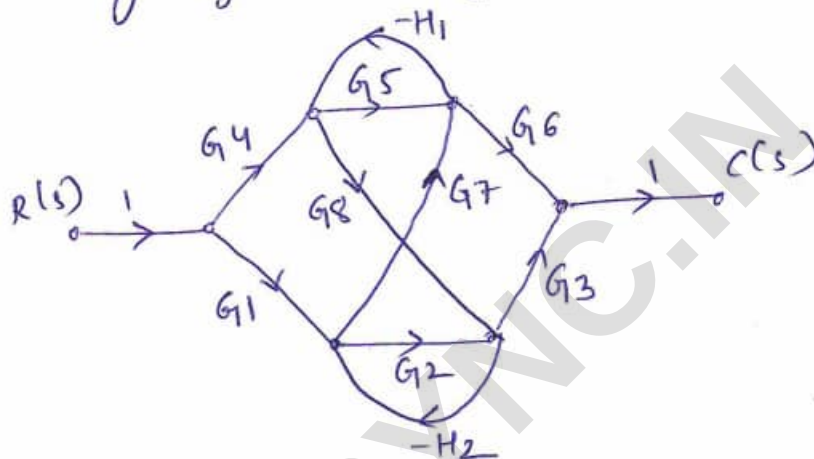
For T_2 , L_1 is non-touching

$$\therefore \Delta_2 = 1 - L_1 = \underline{1 + G_6 H_6}$$



$$\therefore T.F = \frac{C(s)}{R(s)} = \frac{G_1 G_5 G_6 G_7 G_8 (1 + G_3 H_3) + G_1 G_2 G_3 G_4 G_8 (1 + G_6 H_6)}{1 + G_6 H_6 + G_3 H_3 + G_3 G_6 H_3 H_6}$$

③ Using Mason's gain formula, find the gain of the following system in figure below.



Solⁿ: The number of forward paths $\Rightarrow K = 6$

$$\text{Mason's gain formula, } T.F = \frac{C(s)}{R(s)} = \sum_{k=1}^6 \frac{T_k \Delta_k}{\Delta}$$

$$\frac{C(s)}{R(s)} = \frac{T_1 \Delta_1 + T_2 \Delta_2 + T_3 \Delta_3 + T_4 \Delta_4 + T_5 \Delta_5 + T_6 \Delta_6}{\Delta}$$

The forward path gains are,

$$T_1 = G_1 G_2 G_3, T_2 = G_4 G_5 G_6, T_3 = G_1 G_7 G_6$$

$$T_4 = G_4 G_8 G_3, T_5 = G_1 G_7 (-H_1) G_8 G_3$$

$$T_6 = G_4 G_8 (-H_2) G_7 G_6$$

The individual feedback loops and their gains are,

$$L_1 = -G_5 H_1, L_2 = -G_2 H_2 \text{ and } L_3 = G_7 G_8 H_1 H_2$$

The two non-touching loops are L_1 and L_2

$$\therefore L_1 L_2 = G_2 G_5 H_1 H_2$$

$$\therefore \Delta = 1 - [L_1 + L_2 + L_3] + [L_1 L_2]$$

$$= 1 + [G_5 H_1 + G_2 H_2 - G_7 G_8 H_1 H_2] + G_2 G_5 H_1 H_2$$

For T_1 , L_1 is non-touching.

$$\therefore \Delta_1 = 1 - L_1 = 1 + G_5 H_1$$

For T_2 , L_2 is non-touching.

$$\therefore \Delta_2 = 1 - L_2 = 1 + G_2 H_2$$

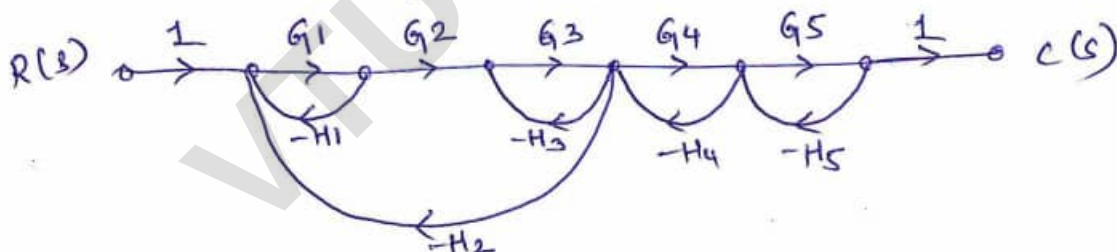
For T_3 to T_6 all loops are touching

$$\therefore \Delta_3 = \Delta_4 = \Delta_5 = \Delta_6 = 1$$

$$\therefore \text{Gain} = T.F = \frac{\sum T_k \cdot \Delta_k}{\Delta} = \frac{T_1 \Delta_1 + T_2 \Delta_2 + T_3 \Delta_3 + T_4 \Delta_4 + T_5 \Delta_5 + T_6 \Delta_6}{\Delta}$$

$$\therefore \text{Gain} = \frac{G_1 G_2 G_3 (1 + G_5 H_1) + G_4 G_5 G_6 (1 + G_2 H_2) + G_1 G_2 G_6 + G_4 G_3 G_3 - G_1 G_3 G_7 G_8 H_1 - G_4 G_6 G_7 G_8 H_2}{1 + G_5 H_1 + G_2 H_2 - G_7 G_8 H_1 H_2 + G_2 G_5 H_1 H_2}$$

(4) Find $C(s)/R(s)$ of signal flow graph:

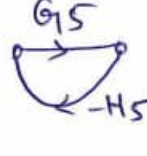
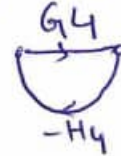
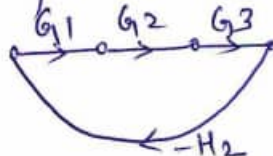
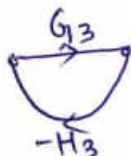


Solⁿ: Number of forward paths $\Rightarrow K = 1$

$$\text{Mason's gain formula, } T.F = \frac{\sum T_k \cdot \Delta_k}{\Delta} = \frac{T_1 \cdot \Delta_1}{\Delta}$$

$$\therefore T_1 = G_1 G_2 G_3 G_4 G_5$$

Individual feedback loops and their gains are,



$$L_1 = -G_1 H_1$$

$$L_2 = -G_2 H_2$$

$$L_3 = -G_1 G_2 G_3 H_2$$

$$L_4 = -G_4 H_4$$

$$L_5 = -G_5 H_5$$

Combinations of two non-touching loops :

- i) L_1 and L_2 ii) L_1 and L_4 iii) L_1 and L_5
 iv) L_2 and L_5 v) L_3 and L_5

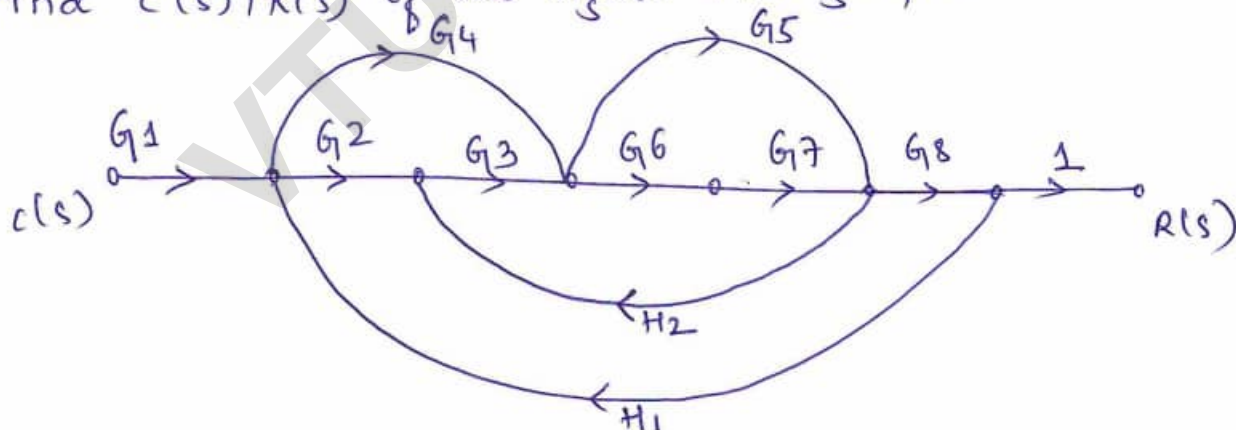
Combination of three non-touching loops: L_1, L_2 and L_5

$$\therefore \Delta = 1 - [L_1 + L_2 + L_3 + L_4 + L_5] + [L_1 L_2 + L_1 L_4 + L_1 L_5 + L_2 L_5 + L_3 L_5] - [L_1 L_2 L_5]$$

$$\Delta = 1 + G_1 H_1 + G_3 H_3 + G_1 G_2 G_3 H_2 + G_4 H_4 + G_5 H_5 + G_1 G_3 H_1 H_3 + G_1 G_4 H_1 H_4 + G_1 G_5 H_1 H_5 + G_3 G_5 H_3 H_5 + G_1 G_2 G_3 G_5 H_2 H_5 + G_1 G_3 G_5 H_1 H_3 H_5$$

$$\therefore \frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4 G_5}{1 + G_1 H_1 + G_3 H_3 + G_1 G_2 G_3 H_2 + G_4 H_4 + G_5 H_5 + G_1 G_3 H_1 H_3 + G_1 G_4 H_1 H_4 + G_1 G_5 H_1 H_5 + G_3 G_5 H_3 H_5 + G_1 G_2 G_3 G_5 H_2 H_5 + G_1 G_3 G_5 H_1 H_3 H_5}$$

⑤ Find $C(s)/R(s)$ of the signal flow graph:



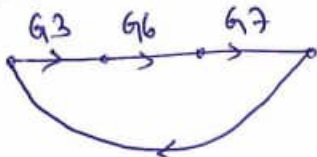
Solⁿ: Number of forward paths $\Rightarrow k = 4$

$$\therefore \text{Mason's Gain formula, T.F} = \frac{\sum T_k \Delta_k}{\Delta} = \frac{T_1 \Delta_1 + T_2 \Delta_2 + T_3 \Delta_3 + T_4 \Delta_4}{\Delta}$$

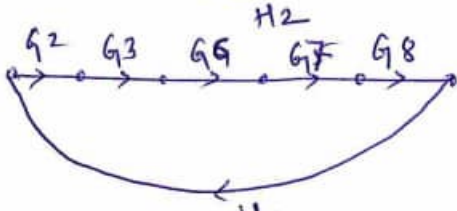
$$\therefore T_1 = G_1 G_2 G_3 G_6 G_7 G_8, \quad T_2 = G_1 G_4 G_6 G_7 G_8$$

$$T_3 = G_1 G_2 G_3 G_5 G_8, \quad T_4 = G_1 G_4 G_5 G_8$$

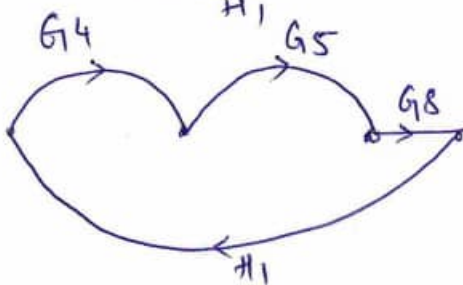
Individual feedback loops and their gains are:



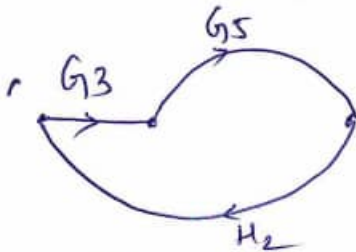
$$L_1 = G_3 G_6 G_7 H_2$$



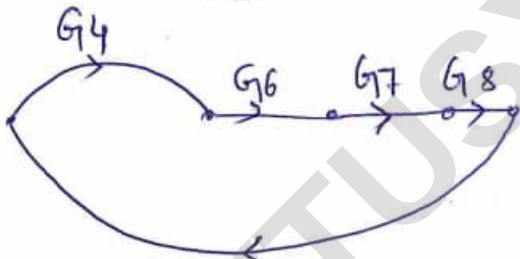
$$L_2 = G_2 G_3 G_6 G_7 G_8 H_1$$



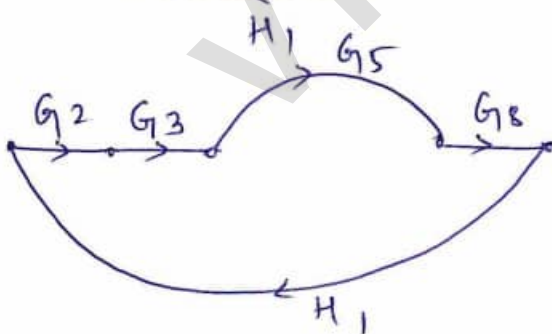
$$L_3 = G_4 G_5 G_8 H_1$$



$$L_4 = G_3 G_5 H_2$$



$$L_5 = G_4 G_6 G_7 G_8 H_1$$



$$L_6 = G_2 G_3 G_5 G_8 H_1$$

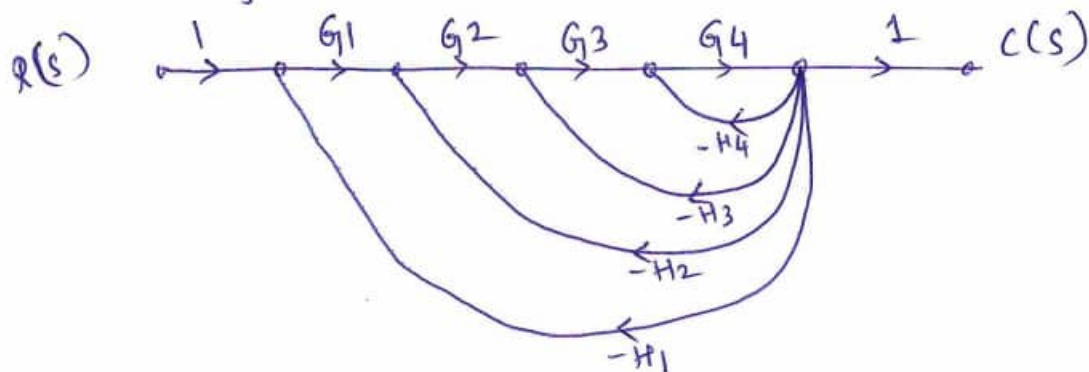
There are no two non-touching loops. $\therefore \Delta = 1 - [L_1 + L_2 + L_3 + L_4 + L_5 + L_6]$

To find Δ_1 to Δ_4 : For T_1 to T_4 all loops are touching

$$\therefore \Delta_1 = \Delta_2 = \Delta_3 = \Delta_4 = 1$$

$$\therefore \frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_6 G_7 G_8 + G_1 G_4 G_6 G_7 G_8 + G_1 G_2 G_3 G_5 G_8 + G_1 G_4 G_5 G_8}{1 - G_3 G_6 G_7 H_2 - G_2 G_3 G_6 G_7 G_8 H_1 - G_4 G_5 G_8 H_1 - G_3 G_5 H_2 - G_4 G_6 G_7 G_8 H_1 - G_2 G_3 G_5 G_8 H_1}$$

⑥ Find $C(s)/R(s)$ of the signal flow graph using Mason's gain formula.

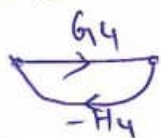


Solⁿ: Number of forward paths, $K = 1$

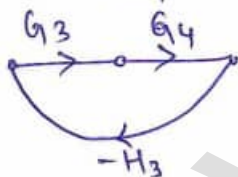
Mason's Gain formula, $T.F = \frac{C(s)}{R(s)} = \frac{\sum T_k \Delta_k}{\Delta}$
 $= \frac{T_1 \Delta_1}{\Delta}$

$\therefore T_1 = G_1 G_2 G_3 G_4$

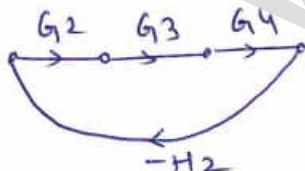
Individual feedback loops and their gains are:



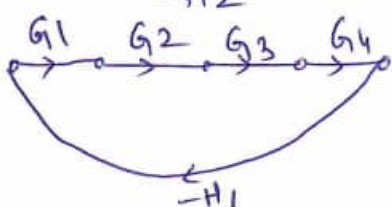
$L_1 = -G_4 H_4$



$L_2 = -G_3 G_4 H_3$



$L_3 = -G_2 G_3 G_4 H_2$



$L_4 = -G_1 G_2 G_3 G_4 H_1$

There are no combinations of two non-touching loops.
 $\Delta = 1 - [L_1 + L_2 + L_3 + L_4]$

To find Δ_1

For T_1 all loops are touching. $\therefore \Delta_1 = 1$

$\therefore \frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4}{1 + G_4 H_4 + G_3 G_4 H_3 + G_2 G_3 G_4 H_2 + G_1 G_2 G_3 G_4 H_1}$

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